

THE DISTRIBUTION OF THE LARGEST NEW EIGENVALUES OF RANDOM COVERS OF GRAPHS

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ABSTRACT. A finite d -regular graph G on V vertices is said to be *one-sided Ramanujan* if its second largest adjacency eigenvalue $\lambda_2(G)$ is at most $2\sqrt{d-1}$, the spectral radius of the infinite d -regular tree. Miller, Novikoff, and Sabelli [32] conjectured that the probability of being Ramanujan tends to approximately 52%. Analyzing 1.3×10^8 samples of the largest new eigenvalue across families of random regular graphs and random covers, we find that the 52% was a finite size artifact of power law fits over small sizes, and that

$$\lim_{V \rightarrow \infty} P(G \text{ is one-sided Ramanujan}) = F_1(0) = 0.8319081\dots,$$

where F_1 is the Tracy-Widom distribution of the Gaussian Orthogonal Ensemble in its original normalization, since proven by Huang, McKenzie, and Yau. Our main experimental finding concerns random covers of graphs: the largest new eigenvalue of a random cover fluctuates according to the Tracy-Widom law of symmetry class $\beta \in \{1, 2, 4\}$ selected by the Frobenius-Schur type of the associated irreducible representation of the covering group, with centering at the spectral radius of the universal cover of the base graph, and the limiting probability that a random cover is Ramanujan is an explicit product of Tracy-Widom functionals, which the data verifies to three decimal places.

1. INTRODUCTION

Let G be a d -regular graph on V vertices with adjacency matrix eigenvalues $d = \lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_V$. The spectral gap $d - \lambda_2$ controls the expansion of G , and the extent to which it can be large is governed by the spectral radius $2\sqrt{d-1}$ of the infinite d -regular tree. Alon and Boppana (see Nilli [33]) proved that for every d -regular graph on V vertices

$$\lambda_2(G) \geq 2\sqrt{d-1} - \frac{C_d}{\log V}$$

for a constant $C_d > 0$, and so $2\sqrt{d-1}$ is the best spectral gap achievable by an infinite family. Following Lubotzky, Phillips, and Sarnak [28], a d -regular graph is called *Ramanujan* if every eigenvalue other than $\pm d$ is at most $2\sqrt{d-1}$ in absolute value, and *one-sided Ramanujan* if

$$\lambda_2(G) \leq 2\sqrt{d-1}.$$

Lubotzky, Phillips, and Sarnak [28] and independently Margulis [30] constructed infinite families of Ramanujan graphs when $d-1$ is prime, using deep results from number theory, and Marcus, Spielman, and Srivastava [29] proved that bipartite Ramanujan graphs exist for every degree. Whether non-bipartite Ramanujan graphs exist for every degree remains open. In a different direction, Alon [1] conjectured that random d -regular graphs are nearly Ramanujan, that is $\lambda_2 \leq 2\sqrt{d-1} + \epsilon$ asymptotically almost surely for every $\epsilon > 0$, and this was proven by Friedman [16].

Friedman's theorem localizes λ_2 at the Ramanujan threshold but says nothing about which side of the threshold it falls on, and Sarnak [35] raised the question of whether a positive proportion of random d -regular graphs are Ramanujan, reporting early experiments of Novikoff and already pointing to the Tracy-Widom law. The first serious numerical attack on this question was made in 2008 by Miller, Novikoff, and Sabelli [32], hereafter MNS. Sampling 1000 graphs per size from bipartite and perfect matching models of d -regular graphs with $d \in \{3, 4, 7, 10\}$ and $V \leq 20,000$, with 100,000 graphs at three selected sizes, and motivated by the numerical evidence of Gaussian Orthogonal Ensemble spacing statistics in the bulk found by Jakobson, Miller, Rivin, and Rudnick [25], they gave evidence that the fluctuations of λ_2 follow the $\beta = 1$ Tracy-Widom law of random matrix theory [38], the law of the largest eigenvalue of the Gaussian Orthogonal Ensemble. Writing μ_V and σ_V for the mean and standard deviation of λ_2 over graphs of size V , they fitted power laws

$$2\sqrt{d-1} - \mu_V \sim c_\mu V^m, \quad \sigma_V \sim c_\sigma V^s,$$

and found best fit exponents m between -0.79 and -0.83 against s between -0.71 and -0.73 for their $d = 3$ families. Since their fitted m was steeper than their fitted s , the threshold $2\sqrt{d-1}$ sits asymptotically at the mean of the distribution, and they were led to conjecture that the limiting probability of being one-sided Ramanujan is the mass of the Tracy-Widom law to the left of its mean,

$$(1.1) \quad F_1(\mu_{TW_1}) = 0.519652\dots,$$

approximately 52%, with $27\% \approx 52\%^2$ of non-bipartite graphs Ramanujan on both sides.

The fluctuation question for random regular graphs has since been answered. For degrees growing with the number of vertices, Bauerschmidt, Huang, Knowles, and Yau [6] proved Tracy-Widom fluctuations for $V^{2/9+\epsilon} \leq d \leq V^{1/3-\epsilon}$, He [20] for $V^{2/3+\epsilon} \leq d \leq V/2$, and Huang and Yau [23] for the full sparse regime $V^\epsilon \leq d \leq V^{1/3-\epsilon}$, concluding that about 69% of such graphs are Ramanujan, and correcting the 27% prediction: the correct one-sided constant is 83%, not 52%. For fixed degree, where Huang and Yau [24] had located λ_2 within V^{-c} of the threshold, Huang, McKenzie, and Yau [22] proved the definitive result.

Theorem 1. *Let $d \geq 3$ be fixed, let G be a uniformly random simple d -regular graph on V vertices, and let $A = d(d-1)/(d-2)^2$. Then*

$$(AV)^{2/3} \left(\frac{\lambda_2(G)}{\sqrt{d-1}} - 2 \right) \rightarrow TW_1$$

in distribution as $V \rightarrow \infty$, where TW_1 is the $\beta = 1$ Tracy-Widom distribution in its original normalization. In particular, since the centering is exactly $2\sqrt{d-1}$,

$$\lim_{V \rightarrow \infty} P \left(\lambda_2(G) \leq 2\sqrt{d-1} \right) = F_1(0) = 0.8319081\dots$$

Unfolding the constants, Theorem 1 states that the mean and standard deviation of λ_2 satisfy

$$(1.2) \quad 2\sqrt{d-1} - \mu_V \approx |\mu_{TW_1}| c(d) V^{-2/3}, \quad \sigma_V \approx \sigma_{TW_1} c(d) V^{-2/3},$$

where $c(d) = \left(\frac{(d-2)^2}{d(d-1)}\right)^{2/3} \sqrt{d-1}$, and where $\mu_{TW_1} = -1.2065335746\dots$ and $\sigma_{TW_1} = 1.2679830577\dots$ are the mean and standard deviation of the un-recentered Tracy-Widom law, so that both of MNS's exponents equal $-2/3$ and the threshold sits $|\mu_{TW_1}|/\sigma_{TW_1} = 0.951537\dots$ standard deviations to the right of the mean.

We report on a numerical study, roughly a thousand times larger than that of MNS, of the largest new eigenvalue of random covers of graphs, a setting which contains the random regular graph as its simplest case and in which, outside one result on 2-covers of dense random regular graphs [26], the fluctuation theory is open. A degree k cover (or k -lift) H of a base graph B is obtained by assigning to each edge of B a permutation in S_k , and the eigenvalues of H split into the *old* eigenvalues, which are those of B , and the remaining $(k-1)|V(B)|$ *new* eigenvalues. Random covers were introduced as a combinatorial construction of expanders by Amit and Linial [4], and Bilu and Linial [7] conjectured that every d -regular graph has a 2-cover whose new eigenvalues all lie in $[-2\sqrt{d-1}, 2\sqrt{d-1}]$, which would produce Ramanujan graphs of every degree. Marcus, Spielman, and Srivastava [29] proved the one-sided statement for bipartite graphs, and Hall, Puder, and Sawin [19] proved that every connected loopless graph has a one-sided Ramanujan cover of every degree. For random covers, Friedman [15] formulated the relative Alon conjecture, that the largest new eigenvalue of a random degree k cover of B is at most $\rho(B) + \epsilon$ asymptotically almost surely, where $\rho(B)$ is the spectral radius of the universal cover of B . Successive bounds were given by Lubetzky, Sudakov, and Vu [27] and by Puder [34], and the conjecture was proven for regular base graphs by Friedman and Kohler [17] and in general by Bordenave [8] (see also [9]). Accordingly, we call a cover H of B *one-sided Ramanujan* if every new eigenvalue is at most $\rho(B)$. When B is d -regular the universal cover is the d -regular tree and $\rho(B) = 2\sqrt{d-1}$, and when B is the bouquet of $d/2$ loops on a single vertex, a random degree k cover of B is precisely the permutation model of a random d -regular graph on k vertices, with new eigenvalues its non-trivial eigenvalues. The relative Alon conjecture is thus the exact analog of Friedman's theorem, and the question answered by Theorem 1 returns in general form: how does the largest new eigenvalue of a random cover fluctuate around $\rho(B)$, and what proportion of random covers are Ramanujan?

Our data supports a precise and, to our knowledge, new answer: the limiting fluctuations follow the Tracy-Widom law of the random matrix symmetry class dictated by the representation theory of the cover. When the covering permutations are constrained to a group Γ , the new spectrum decomposes into isotypic blocks indexed by the non-trivial irreducible representations of Γ (see section 2), and the Frobenius-Schur indicator of each representation sorts its block into one of the three classical ensembles: real representations give real symmetric blocks (GOE, $\beta = 1$), complex representations give complex Hermitian blocks (GUE, $\beta = 2$), and quaternionic representations give quaternion self-dual blocks (GSE, $\beta = 4$), the threefold way of Dyson [13]. Our main conjecture is that the block fluctuations realize this trichotomy.

Conjecture 2. *Fix $d \geq 3$ and a finite group Γ , let ρ be a non-trivial irreducible representation of Γ with Frobenius-Schur indicator $\varepsilon(\rho) \in \{1, 0, -1\}$, and let A_ρ denote the ρ -isotypic block of a uniformly random Γ -cover of a random simple d -regular base graph on n vertices. Then there is a constant $c_\rho > 0$ such that*

$$\frac{\lambda_{\max}(A_\rho) - 2\sqrt{d-1}}{c_\rho n^{-2/3}} \rightarrow TW_\beta$$

in distribution as $n \rightarrow \infty$, where $\beta = 1, 2, 4$ according as $\varepsilon(\rho) = 1, 0, -1$, and TW_β is the Tracy-Widom law of the corresponding ensemble in its original normalization. In particular the centering is exactly $2\sqrt{d-1}$, with no shift.

The largest new eigenvalue of the cover is the maximum of the block maxima, and blocks of representations that are not complex conjugates of one another behave independently in our data. This yields explicit limiting Ramanujan probabilities.

Conjecture 3. *With the notation of Conjecture 2, the block maxima $\lambda_{\max}(A_\rho)$ are asymptotically independent across non-conjugate non-trivial representations, and the probability that a random Γ -cover is one-sided Ramanujan satisfies*

$$\lim_{n \rightarrow \infty} P\left(\lambda_{\max}(A_\rho) \leq 2\sqrt{d-1} \text{ for all } \rho\right) = \prod_{[\rho] \neq 1} F_{\beta(\rho)}(0),$$

where the product runs over the non-trivial irreducible representations of Γ up to complex conjugation, and F_β is the Tracy-Widom distribution function. Here $F_1(0) = 0.8319081\dots$, $F_2(0) = 0.96937\dots$, and $F_4(0) = 0.99857\dots$.

When all the non-trivial blocks share one symmetry class the largest new eigenvalue itself is Tracy-Widom, and this is what the data shows: $\mathbb{Z}_3$ covers follow TW_2 , quaternion covers follow TW_4 , and permutation covers, whose new spectrum is the single block of the real standard representation of S_k , follow TW_1 . When the blocks mix classes no single Tracy-Widom law can appear, and the data confirms this too: the largest new eigenvalue of a $\mathbb{Z}_4$ cover, a maximum of one GOE block and one GUE block, and of a $\mathbb{Z}_5$ cover, a maximum of two independent GUE blocks, each drift away from every Tracy-Widom law, while matching the predicted maximum law to three decimal places (see Table 1 in section 4). Our final conjecture extends the picture to covers of a fixed irregular base graph, where no theorem and, to our knowledge, no prior experiment exists. For regular bases, Lubetzky, Sudakov, and Vu [27] simulated random covers of three cubic Ramanujan graphs and conjectured that the probability that a random cover is Ramanujan converges to a constant in $(0, 1)$, and the following conjecture identifies their constant, universally in the base graph.

Conjecture 4. *Let B be a fixed connected base graph of minimum degree at least 3, let $\rho(B)$ denote the spectral radius of its universal cover, and let H_k be a uniformly random degree k cover of B . Then there is a constant $c_B > 0$ such that*

$$\frac{\lambda_{\text{new}}(H_k) - \rho(B)}{c_B k^{-2/3}} \rightarrow TW_1$$

in distribution as $k \rightarrow \infty$, where λ_{new} is the largest new eigenvalue. In particular

$$\lim_{k \rightarrow \infty} P(H_k \text{ is one-sided Ramanujan}) = F_1(0) = 0.8319081\dots$$

for every such base graph B .

The $\beta = 1$ law in Conjecture 4 is forced by the picture above, since the standard representation of S_k is real for every k . Taking B to be the bouquet of two loops recovers the $d = 4$ case of Theorem 1 (in the permutation model), and so Conjecture 4 is the common generalization of the proven regular case to arbitrary base graphs. Both cases are tested in section 5 at five million covers per size of the fixed bases K_5 and $K_5 - e$: each family selects the TW_1 law, and the empirical Ramanujan probabilities at $V = 10,000$ are 0.83281 and 0.82916 against the conjectured $F_1(0) = 0.8319081\dots$.

The conjectures above date to 2018 and were formulated jointly with Lior Silberman, following a suggestion of Peter Sarnak. All of the data reported here was generated by 2023, before the appearance of [22], with every family but the two fixed base ones predating [23] as well, and so the pipeline was in no way tuned to the theorems it now confirms. This timing shapes how the paper should be read: the regular graph families of section 3 are no longer evidence for a conjecture but a control experiment, and the value of that section lies in what the theorem does not address, namely the finite size behavior, the diagnosis of MNS's 52%, and the failure of the multigraph model. The cover families of sections 4 and 5 are the substance of the paper.

We stress three differences between our experiment and that of MNS. First, scale: we use 5×10^6 samples per size against their 1000, sizes up to $V = 500,000$ against their 20,000, and 1.3×10^8 samples in total, so that the four candidate laws, whose distribution functions differ only in the third decimal, are separated by dozens of binomial standard errors rather than fractions of one. Second, methodology: instead of fitting a single power law across all sizes we track the local, consecutive size exponents of the mean gap and standard deviation, which reveals that a global fit over $V \leq 20,000$ systematically overestimates the mean exponent, the precise finite size artifact behind (1.1). Third, and principally, scope: the nearest precursor to our cover experiments is the simulation of [27], one thousand covers per size of three cubic Ramanujan bases, which observed a stable Ramanujan proportion without identifying a law. The representation theoretic selection of the symmetry class, the Tracy-Widom identification, and the explicit constants have, to our knowledge, no analog in any published experiment. Throughout, we study the one-sided question only, and make no claims about the joint behavior of the smallest new eigenvalue.

1.1. Outline. In section 2 we describe the random cover model, the isotypic decomposition, the statistics we compute, and the dataset. In section 3 we present the regular graph control experiment: the convergence to the proven law (1.2), the local exponent diagnosis of MNS's conjecture, and the failure of the permutation multigraph model. Section 4 presents the evidence for Conjectures 2 and 3 across abelian and quaternion covers, together with a quantitative account of the mixed class families. Section 5 presents Conjecture 4 and the first data for covers of K_5 minus an edge. Section 6 collects open problems.

2. THE NUMERICAL MODEL AND THE DATASET

2.1. Random Covers and the Isotypic Decomposition. Let B be a connected base graph on n vertices, possibly with loops and multiple edges. A degree k cover of B is determined by a voltage assignment σ giving each oriented edge $e = (u, v)$ of B a permutation $\sigma_e \in S_k$, with $\sigma_{\bar{e}} = \sigma_e^{-1}$ on the reversed edge: the cover H has vertex set $V(B) \times \{1, \dots, k\}$, with (u, i) adjacent to $(v, \sigma_e(i))$ for each edge $e = (u, v)$. A *random cover* of degree k assigns independent uniform permutations to the edges of B . More generally, given a finite group Γ of order k , a random Γ -cover assigns independent uniform elements of Γ to the edges, acting on the fiber Γ by right multiplication. The eigenvalues of B lift to H , and are called the old eigenvalues, and the remaining $n(k-1)$ eigenvalues are called *new*, and we write $\lambda_{\text{new}}(H)$ for the largest new eigenvalue.

The spectrum of a Γ -cover organizes along the irreducible representations of Γ . Decomposing the permutation action of Γ on its group algebra into irreducibles, the adjacency matrix of H is conjugate to a direct sum

$$(2.1) \quad A_H \cong \bigoplus_{\rho \in \text{Irr}(\Gamma)} A_\rho^{\oplus d_\rho}, \quad A_\rho = \sum_{e=(u,v)} E_{uv} \otimes \rho(\sigma_e),$$

where the sum runs over the oriented edges of B , E_{uv} is the matrix unit, and d_ρ is the dimension of ρ , so that A_ρ is an nd_ρ dimensional Hermitian matrix (see [19, Section 2] for this decomposition). The trivial representation recovers the base, $A_1 = A_B$, and so the new spectrum of H is the union of the spectra of the non-trivial blocks A_ρ , each with multiplicity d_ρ . Note that the blocks of complex conjugate representations satisfy $A_{\bar{\rho}} = \overline{A_\rho}$ and have identical spectra, and so the distinct sources of randomness are indexed by the non-trivial representations up to conjugation. For a full symmetric cover the same decomposition applies with $\Gamma = S_k$ acting on $\{1, \dots, k\}$: the permutation representation splits as the trivial plus the standard $(k-1)$ dimensional representation, and the new spectrum is the single block of the standard representation.

The Frobenius-Schur indicator

$$\varepsilon(\rho) = \frac{1}{|\Gamma|} \sum_{g \in \Gamma} \chi_\rho(g^2) \in \{1, 0, -1\}$$

sorts the irreducible representations into real ($\varepsilon = 1$), complex ($\varepsilon = 0$), and quaternionic ($\varepsilon = -1$) types, and correspondingly sorts the blocks A_ρ into the three classical matrix symmetry classes: a real representation may be realized over $\mathbb{R}$, making A_ρ conjugate to a real symmetric matrix, a complex representation gives a complex Hermitian block with no further symmetry, and a quaternionic representation gives a self-dual block whose eigenvalues carry an even multiplicity. In random matrix theory these are the symmetry classes of the GOE, GUE, and GSE, whose largest eigenvalues fluctuate according to the Tracy-Widom laws TW_1 , TW_2 , and TW_4 [37, 38]. This threefold way [13] is the mechanism behind Conjecture 2. Our families realize all three types: for $\Gamma = \mathbb{Z}_k$ the non-trivial characters $\chi_j(x) = e^{2\pi i j x/k}$ give n dimensional blocks with entries $\chi_j(\sigma_e)$ on the edges of the base, complex Hermitian in general and real symmetric exactly when $\chi_j = \pm 1$ is real valued, while the quaternion group Q_8 has a unique non-trivial higher dimensional representation, the two dimensional representation of quaternionic type, whose real form is four dimensional.

Concretely, $\mathbb{Z}_3$ has one conjugate pair of complex characters, and so the largest new eigenvalue of a random $\mathbb{Z}_3$ cover is the largest eigenvalue of a single GUE class block. $\mathbb{Z}_4$ has the real character $\chi_2 = (-1)^x$ together with one conjugate pair, and so its largest new eigenvalue is the maximum of a GOE class and a GUE class maximum. $\mathbb{Z}_5$ has two conjugate pairs, giving the maximum of two GUE class maxima with identical marginal distributions. In general, the non-trivial characters of $\mathbb{Z}_k$ form $\lfloor (k-1)/2 \rfloor$ conjugate pairs, together with the real character $\chi_{k/2}(x) = (-1)^x$ when k is even, and so the largest new eigenvalue of a random cyclic cover is the maximum of $\lfloor (k-1)/2 \rfloor$ independent GUE class maxima and, for even k , one GOE class maximum, with limiting Ramanujan probability

$$F_2(0)^{\frac{k-1}{2}} \quad \text{for odd } k, \quad F_1(0) F_2(0)^{\frac{k-2}{2}} \quad \text{for even } k$$

under Conjecture 3. The quaternion family samples the four dimensional real form of the quaternionic block of a random Q_8 -cover directly, a $4n$ dimensional matrix all of whose eigenvalues are new, with each eigenvalue doubled by the quaternionic structure.

These five families, together with the permutation covers of the bouquet (random 4-regular graphs), of K_5 , and of K_5 minus an edge, are the subject of sections 3 through 5.

2.2. The Statistics and the Tracy-Widom Constants. By TW_β we mean the Tracy-Widom law in the original normalization of Tracy and Widom [37, 38], with distribution function F_β , mean μ_β , and standard deviation σ_β , and by $\widetilde{TW}_\beta$ the standardization to mean 0 and variance 1. The constants we use are

$$(2.2) \quad \begin{array}{ccccc} \beta & \mu_\beta & \sigma_\beta & |\mu_\beta|/\sigma_\beta & F_\beta(\mu_\beta) \\ 1 & -1.2065335746 & 1.2679830577 & 0.9515376 & 0.519652 \\ 2 & -1.7710868074 & 0.9017731382 & 1.9640048 & 0.515016 \\ 4 & -2.3068848932 & 0.7195302084 & 3.2060987 & 0.511069 \end{array}$$

following [11, 10], and the values $F_1(0) = 0.8319081\dots$, $F_2(0) = 0.96937\dots$, $F_4(0) = 0.99857\dots$ of the distribution functions at the origin.

For each family and size we record the sample mean μ_V , standard deviation σ_V , skewness, and excess kurtosis of the largest new eigenvalue λ , the Kolmogorov-Smirnov distance between the standardized sample and each of $\widetilde{TW}_1, \widetilde{TW}_2, \widetilde{TW}_4$ and the standard normal, the fraction of samples below the Ramanujan threshold, and the fraction below the sample mean. The last statistic is MNS's discriminator: the standardized Tracy-Widom laws are nearly indistinguishable by their overall shape, however their masses to the left of the mean, listed in (2.2), differ in the third decimal, and with 10^6 to 5×10^6 samples the binomial standard error is between 2×10^{-4} and 5×10^{-4} , separating the four candidates by tens of standard errors.

The relevance of the ratio $(2\sqrt{d-1} - \mu_V)/\sigma_V$ is the following elementary observation of MNS. Writing g_V for this ratio, the probability of being one-sided Ramanujan is

$$P\left(\lambda \leq 2\sqrt{d-1}\right) = P\left(\frac{\lambda - \mu_V}{\sigma_V} \leq g_V\right),$$

and so if the standardized law converges to $\widetilde{TW}_1$, then

$$(2.3) \quad \lim_{V \rightarrow \infty} P\left(\lambda \leq 2\sqrt{d-1}\right) = \begin{cases} F_1(\mu_{TW_1}) = 0.519652\dots & \text{if } g_V \rightarrow 0 \\ \widetilde{F}_1(r) & \text{if } g_V \rightarrow r \in (0, \infty) \\ 1 & \text{if } g_V \rightarrow \infty \end{cases}$$

MNS's power law fits pointed to the first case. The Tracy-Widom scaling (1.2), with no shift in the centering, forces the middle case with $r = |\mu_{TW_1}|/\sigma_{TW_1}$, where $\widetilde{F}_1(r) = F_1(0)$, and this is what our data shows and what Theorem 1 now proves.

2.3. The Dataset and the Pipeline. The dataset consists of 1.3×10^8 samples across nine series, each sample the largest new eigenvalue of one independently generated graph or cover, all with base degree $d = 4$.

family	sizes	samples	model
simple 4-regular	$100 \leq V \leq 20,000$ (8)	5,000,000	two permutations, simple
simple 4-regular (legacy)	$5 \times 10^4 \leq V \leq 5 \times 10^5$ (4)	100,000	two permutations, simple
permutation multigraph	$5 \times 10^4 \leq V \leq 5 \times 10^5$ (4)	100,000	loops and multiple edges kept
$\mathbb{Z}_k$ covers, $k = 3, 4, 5$	$100 \leq n \leq 5000$ (6 each)	1,000,000	random simple 4-regular base
quaternion covers	$100 \leq n \leq 5000$ (6)	1,000,000	random simple 4-regular base
covers of K_5	$100 \leq V \leq 10,000$ (7)	5,000,000	fixed base K_5
covers of $K_5 - e$	$100 \leq V \leq 10,000$ (7)	5,000,000	fixed irregular base

The simple 4-regular model draws two independent uniform permutations π_1, π_2 of $\{1, \dots, V\}$ and forms the graph with adjacency matrix $P_1 + P_1^T + P_2 + P_2^T$, resampling locally until the result has no loops or multiple edges, and discarding the rare disconnected samples. This is the permutation model, analyzed first by Friedman [14] (MNS's family $G_{V,4}$), conditioned on simplicity, equivalently a random degree V cover of the bouquet of two loops. Conditioned on simplicity it is contiguous with the uniform simple 4-regular model, see [18] and the survey [39], and so Theorem 1 applies. The permutation multigraph series keeps the loops and multiple edges, with a loop contributing 2 to the diagonal. The abelian series draw a fresh random simple 4-regular base graph on n vertices for each batch, assign each base edge an independent uniform voltage in $\mathbb{Z}_k$, and record the largest new eigenvalue of the explicit kn -vertex cover, ten covers per base graph. A one-way analysis of variance across the ten covers sharing a base gives intraclass correlations below 3×10^{-4} in absolute value at the largest sizes, and below 2.4×10^{-3} at the smallest, and so the residual randomness of the base is invisible at the fluctuation scale and the samples may be treated as independent. The quaternion series assigns each edge of a random simple 4-regular base an independent uniform element of Q_8 and records the top eigenvalue of the resulting $4n$ dimensional real block. The two fixed base series assign each edge of K_5 , respectively of K_5 minus an edge, an independent uniform permutation in S_k , and record the largest new eigenvalue of the cover on $5k$ vertices, with the five lifted base eigenvalues removed exactly. Both families are analyzed in section 5.

Eigenvalues are computed with the implicitly restarted Lanczos method (ARPACK), applied to the adjacency matrix shifted by a small multiple of the identity so that the algebraically largest eigenvalues dominate in magnitude. Old eigenvalues are removed by comparing against the base spectrum, computed exactly, with a tolerance of 10^{-10} . For the regular families this removes the trivial eigenvalue d and yields λ_2 . The number of computed eigenvalues per graph starts at 2 and grows automatically whenever every computed eigenvalue is old. Two precision caveats apply. The legacy series were generated years earlier in MATLAB and are recorded on a 10^{-12} grid, coarser than machine precision. A per process seeding scheme caused seed collisions between worker tasks in the abelian series: of its 1.8×10^6 base graph batches, 452,508 (25%) are redundant copies of another batch, concentrated at the smallest sizes (55 to 81% of batches at $n \in \{100, 200\}$, and below 0.1% at $n \geq 1000$), so that the effective number of independent samples at the smallest sizes is smaller than the nominal 10^6 . Duplicated batches agree to better than 3×10^{-13} , and dropping them moves every statistic quoted in this paper by at most two units in its final displayed digit, consistent with the sampling fluctuation of the removed samples. All summary statistics were verified by independent recomputation from the raw samples.

3. RANDOM 4-REGULAR GRAPHS

In this section we test the pipeline against Theorem 1 along the two simple 4-regular series, diagnose the finite size artifact behind MNS's 52%, and isolate a model, the permutation multigraph, which falls outside the universality class. Since the data predates [23, 22], none of what follows was tuned to the answers.

3.1. Convergence to the Proven Law. Figure 3.1 shows the standardized empirical density of 5×10^6 samples at $V = 20,000$ against the four standardized candidate laws, and Figure 3.2 the quantile-quantile plot of the largest legacy dataset, $V = 500,000$, against $\widetilde{TW}_1$ across the probability range 10^{-4} to 0.999. The Kolmogorov-Smirnov

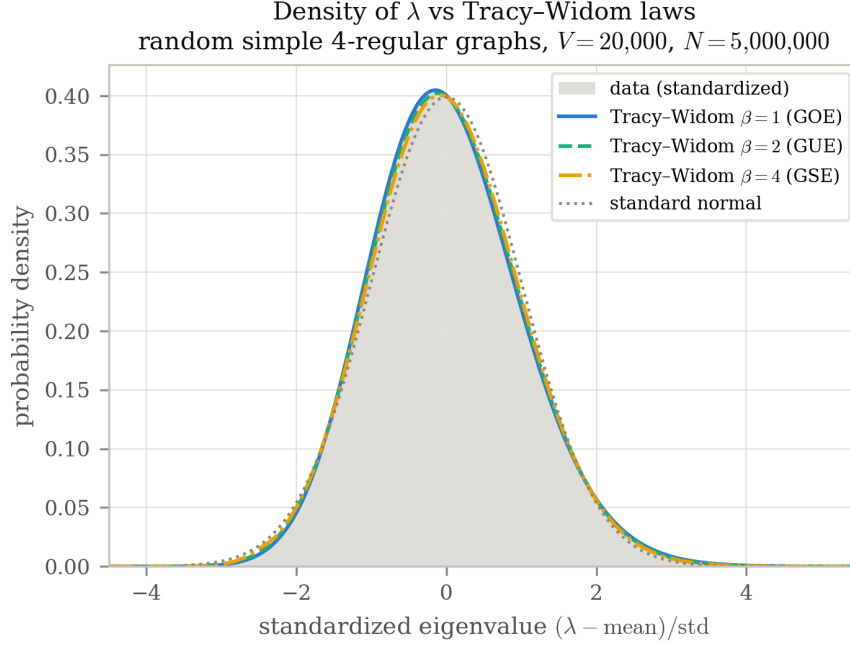


FIGURE 3.1. Standardized empirical density of the largest non-trivial eigenvalue over 5×10^6 random simple 4-regular graphs on $V = 20,000$ vertices, against the standardized Tracy-Widom $\beta = 1, 2, 4$ densities and the standard normal.

distances in Figure 3.3 quantify the picture across all sizes: the distance to $\widetilde{TW}_1$ falls with V to the sampling floor, reaching 8.7×10^{-4} at $V = 20,000$ (the floor at 5×10^6 samples is roughly 4×10^{-4}), while the distances to $\widetilde{TW}_2$, $\widetilde{TW}_4$, and the normal stall at 3.9×10^{-3} , 7.8×10^{-3} , and 1.9×10^{-2} respectively. At the legacy sizes the fit is statistically exact: at $V = 500,000$ the Kolmogorov-Smirnov test accepts $\widetilde{TW}_1$ with $p = 0.99$. The mass to the left of the mean, MNS's decisive statistic, reaches 0.51885 at $V = 20,000$ and 0.52011 at $V = 500,000$ against the TW_1 value 0.519652 from (2.2), with the competing values 0.515016, 0.511069, and 0.5 excluded by more than fifteen binomial standard errors (see Figure 4.3 in section 4). The sample skewness and excess kurtosis at the largest sizes, 0.286 and 0.156 at $V = 20,000$, converge to the TW_1 values 0.293 and 0.165 (Figure 4.4, also in section 4).

The scaling constants of (1.2) are confirmed as well. For $d = 4$ we have that $c(4) = 3^{-1/6} = 0.8326832\dots$, and so the conjectured, now proven, amplitudes are

$$(3.1) \quad 2\sqrt{3} - \mu_V \approx 1.0046602 V^{-2/3}, \quad \sigma_V \approx 1.0558282 V^{-2/3}.$$

Figure 3.4 shows the mean gap and standard deviation against V on logarithmic axes with the $V^{-2/3}$ guide, and Figure 3.5 shows the compensated quantities $(2\sqrt{3} - \mu_V) V^{2/3}$ and $\sigma_V V^{2/3}$, which approach the two constants of (3.1): at $V = 500,000$ the measured standard deviation is 1.685×10^{-4} against the predicted 1.676×10^{-4} , and the measured gap is 1.684×10^{-4} against the predicted 1.595×10^{-4} , the mean converging visibly more slowly than the standard deviation. The constants (3.1) appear in Sodin's work [36] on a closely related model in which the matrix entries carry independent random signs. We note that the arXiv posting of [36] was withdrawn over an error in one of its lemmas,

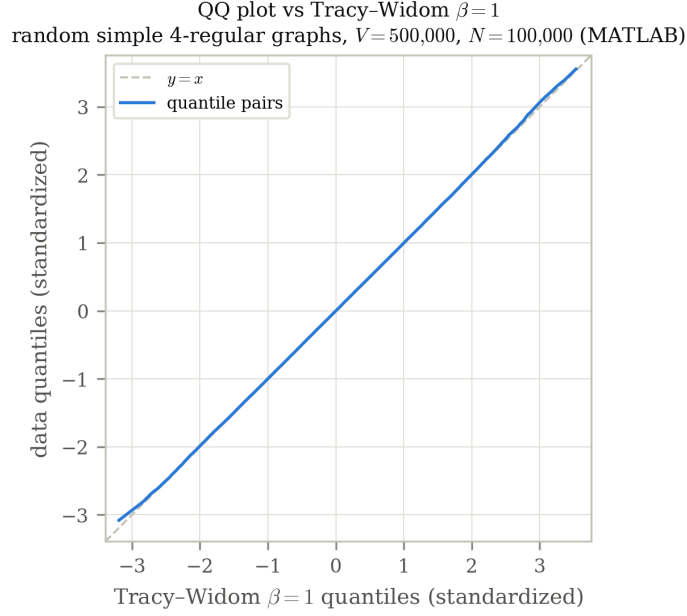


FIGURE 3.2. Quantile-quantile plot of 10^5 samples at $V = 500,000$ against $\widehat{TW}_1$, probability range 10^{-4} to 0.999.

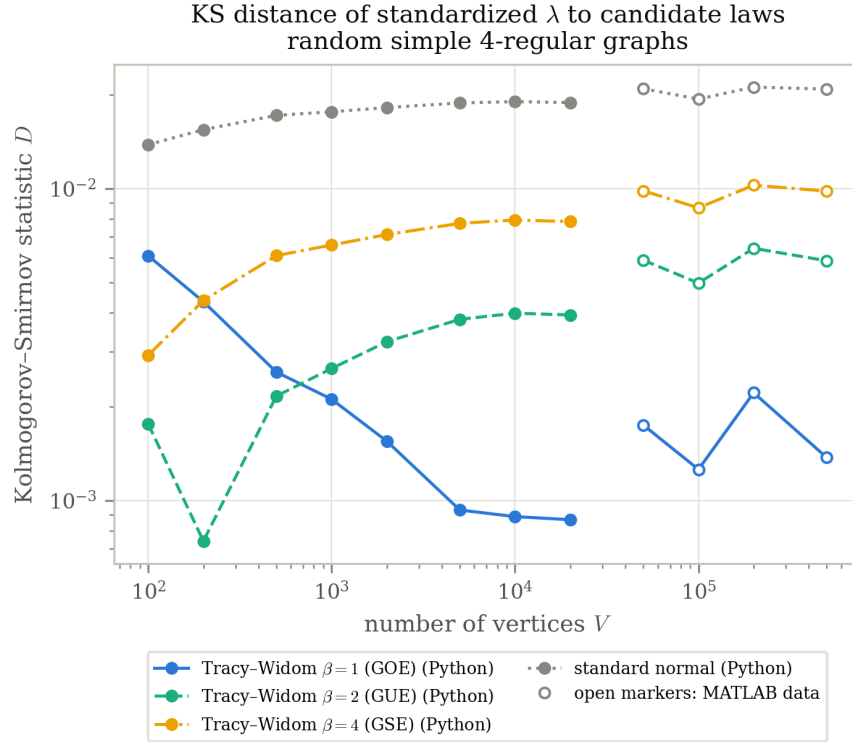


FIGURE 3.3. Kolmogorov-Smirnov distance of the standardized samples to the standardized Tracy-Widom $\beta = 1, 2, 4$ laws and the standard normal, for random simple 4-regular graphs (filled markers: 5×10^6 samples per size, open markers: legacy data with 10^5 samples). Only the distance to $\beta = 1$ keeps falling.

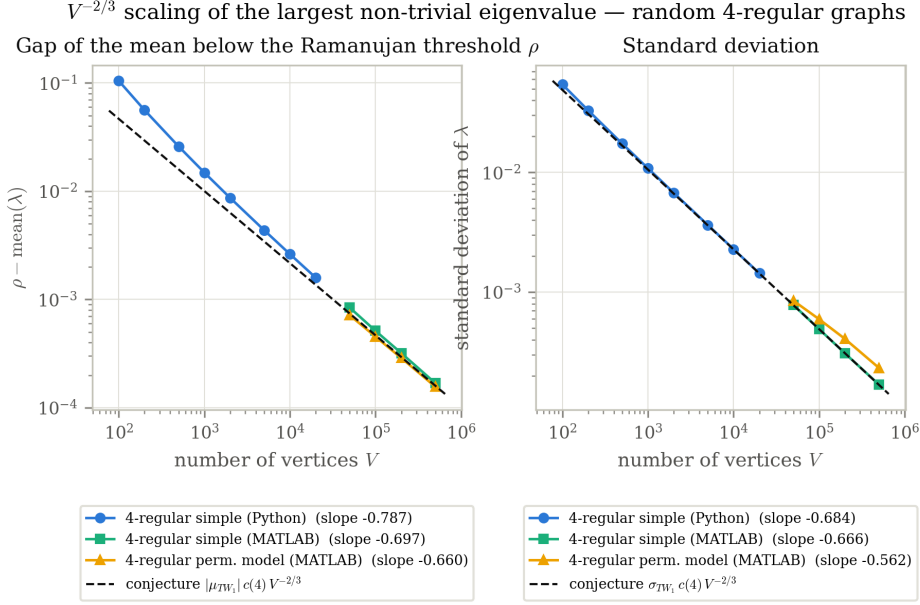


FIGURE 3.4. Mean gap $2\sqrt{3} - \mu_V$ and standard deviation against V for the three 4-regular series, with fitted slopes and the $V^{-2/3}$ Tracy-Widom line.

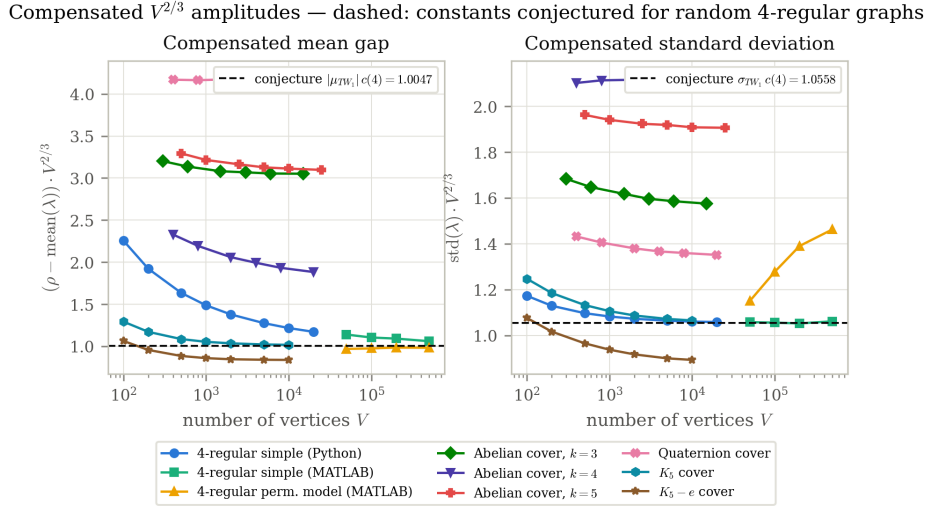


FIGURE 3.5. Compensated scaling: $(\rho - \mu_V) V^{2/3}$ and $\sigma_V V^{2/3}$ against V for all families, where ρ is the family's Ramanujan threshold. The dashed lines are the regular graph amplitudes (3.1), which the simple 4-regular and K_5 cover series approach, while each remaining cover family plateaus at its own amplitude, so the $V^{-2/3}$ rate is universal while the constant is family dependent (see sections 4 and 5).

and the published version carries a 2017 erratum, so that Theorem 1 is the rigorous source for (1.2).

3.2. The Approach to the Limit. The theorems give no rate of convergence, and the data is precise enough to measure one. Write F_V for the distribution function of the

The deviation of the standardized ECDF from TW_1 is a single skewness profile with power-law amplitude

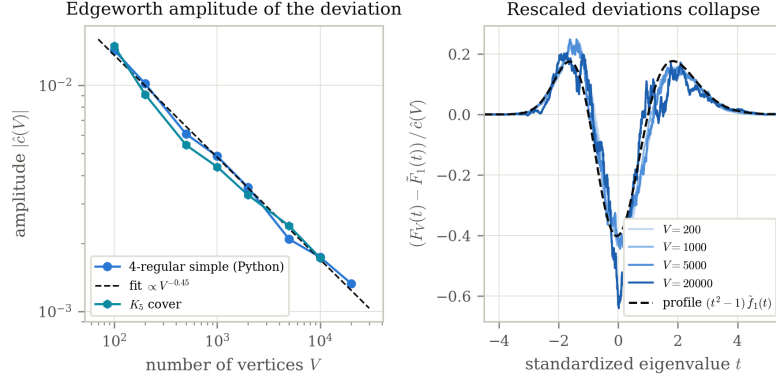


FIGURE 3.6. The approach to the limit. Left: the amplitude $\hat{c}(V)$ of the deviation of the standardized empirical distribution from $\widetilde{TW}_1$, for the simple 4-regular and K_5 cover families, with the fitted power law. Right: the deviations at four sizes, rescaled by $\hat{c}(V)$, collapse onto the fixed profile $(t^2 - 1)\tilde{f}_1(t)$ of a skewness deficit.

standardized largest non-trivial eigenvalue and $\tilde{F}_1, \tilde{f}_1$ for the standardized Tracy-Widom distribution and density. Across the simple 4-regular family the deviation $F_V - \tilde{F}_1$ is, to within 6 to 14 percent of its squared norm, proportional to the single fixed profile $(t^2 - 1)\tilde{f}_1(t)$, the signature of a skewness deficit, and so to the resolution of 5×10^6 samples the finite size law is the one parameter family

$$F_V(t) \approx \tilde{F}_1(t) - \hat{c}(V) (t^2 - 1) \tilde{f}_1(t),$$

where the fitted amplitude $\hat{c}(V)$ decays like a power law with exponent -0.46 ± 0.04 , and the K_5 cover family of section 5 lies on the same curve (Figure 3.6). The same parameter accounts for the residuals of the individual statistics: the mass left of the mean at $V = 20,000$ still sits -8.0×10^{-4} from the TW_1 value, 3.6 binomial standard errors, against an Edgeworth prediction of $-\hat{c}\tilde{f}_1(0) = -5.3 \times 10^{-4}$, and after subtracting the fitted profile the left tail of every 5×10^6 sample dataset agrees with $\widetilde{TW}_1$ to within two standard errors down to probability 6×10^{-7} . The corrections to the amplitudes decay faster: fitting $A + BV^{-\beta}$ to the compensated standard deviation of the simple family gives $A = 1.0548$ with $\beta = 0.63$ freely fitted and $A = 1.0562$ with β fixed at $2/3$, bracketing the conjectured $\sigma_{TW_1}c(4) = 1.05583$ of (3.1) to within a tenth of a percent, while the compensated mean gap admits no single correction exponent over our range, which is why it converges visibly more slowly in Figure 3.5.

3.3. The Origin of the 52% Conjecture. MNS reached the first case of (2.3) by fitting single power laws to the mean gap and standard deviation over their full range of sizes. Our data reproduces their fit and shows exactly how it misleads. Fitting one line to $\log(2\sqrt{3} - \mu_V)$ against $\log V$ over our eight sizes $V \leq 20,000$, the MNS range, gives a slope of -0.787 , while the same fit to $\log \sigma_V$ gives -0.684 : a mean exponent apparently steeper than the standard deviation exponent, exactly the pattern from which MNS concluded $g_V \rightarrow 0$ and hence 52%, with numbers closely matching their $d = 3$ values $m \approx -0.79$ and $s \approx -0.72$. The local exponents tell a different story. Figure 3.7 plots the slope between consecutive sizes: the standard deviation exponent sits within 0.01 of $-2/3$ from $V \approx 2000$ onward, while the mean gap exponent drifts

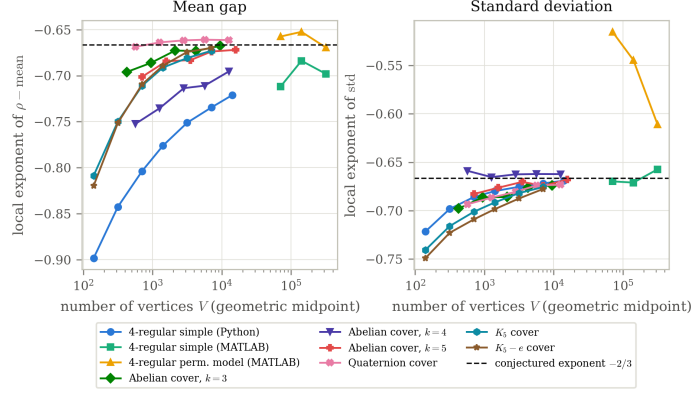
Local log-log slopes between consecutive sizes — a single power-law fit over small V overestimates the gap exponent

FIGURE 3.7. Local (consecutive size) log-log exponents of the mean gap and the standard deviation for all series, with the $-2/3$ line. The mean gap exponent approaches $-2/3$ from below as V grows, and so a single power law fitted over small sizes overestimates its steepness, wrongly suggesting that the gap outruns the standard deviation. This is the origin of the 52% conjecture.

monotonically from -0.90 at $V \approx 100$ through -0.72 at $V = 20,000$ to -0.70 over the last legacy pair at $V = 500,000$, still descending toward $-2/3$. A single fitted line over small sizes averages this drift and reports a transient. MNS saw the warning sign, observing that their best fit exponents increased monotonically as sizes were added, and were appropriately cautious. At $V \leq 20,000$ the data could do no better. The ratio g_V resolves the question directly: as Figure 4.7 shows, g_V does not tend to 0 for any family, and for the simple 4-regular series it descends to 0.9993 at $V = 500,000$, approaching the Tracy-Widom ratio $|\mu_{TW_1}|/\sigma_{TW_1} = 0.951537\dots$ from above, in agreement with the middle case of (2.3) and with $F_1(0) = 0.8319081\dots$ as the limiting Ramanujan probability.

3.4. Failure of the Multigraph Model. MNS pooled models with and without loops and multiple edges, observing similar behavior at their sample sizes. At ours, the difference is qualitative. The permutation multigraph series, identical to the legacy simple series except that loops and multiple edges are kept, has Kolmogorov-Smirnov distance to every candidate law growing with V , reaching 0.085 at $V = 500,000$, sample skewness 74.5 , and excess kurtosis 1.15×10^4 , against the TW_1 values 0.293 and 0.165 . Figure 3.8 shows the mechanism: the right tail of the simple model dies within 10^{-4} of the threshold $2\sqrt{3}$, while the multigraph model has a heavy tail of far outliers, reaching $\lambda = 3.538$ at $V = 50,000$, with roughly 2×10^{-5} of its mass more than ten standard deviations above the threshold at $V = 500,000$ and an extreme sample 155 standard deviations above the mean. These outliers are localized eigenvalues seeded by loops and multiple edges, whose density vanishes in the limit but whose spectral effect does not vanish at the $V^{-2/3}$ fluctuation scale. The lesson for experiments in this area is that the universality class is a property of the conditioned model: the simple and multigraph models share the same limiting spectral distribution and the same almost sure convergence of λ_2 to $2\sqrt{3}$, and are separated only at the fluctuation scale, where one is Tracy-Widom and the other is not even tight around the threshold in units of its own standard deviation, its ratio g_V falling through 0.67 with no sign of stabilizing.

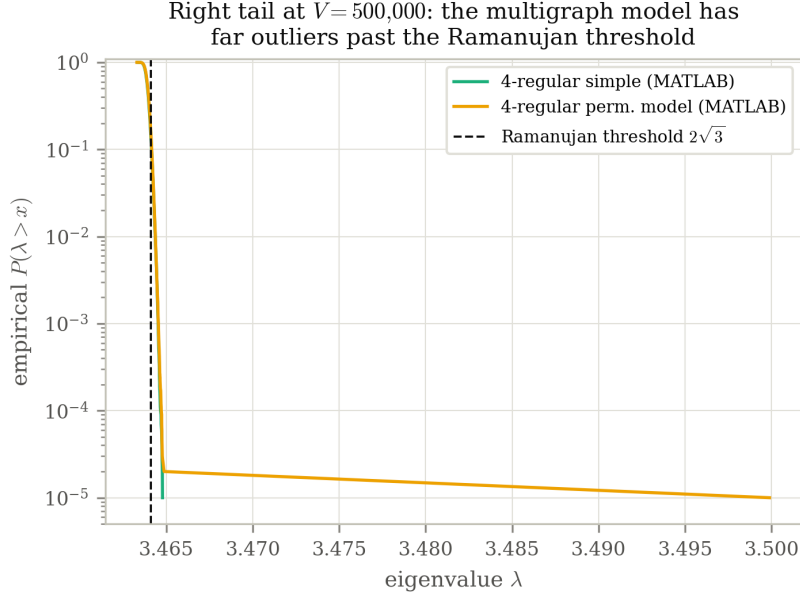


FIGURE 3.8. Right tail $P(\lambda > x)$ at $V = 500,000$ for the simple and multigraph permutation models on logarithmic scale. The simple model's tail terminates within 10^{-4} of $2\sqrt{3}$, while the multigraph model's outliers, localized eigenvalues from loops and multiple edges, reach far beyond the threshold.

4. COVERS AND THE THREEFOLD WAY

We now turn to the main experiment. Each abelian and quaternion family isolates a definite set of isotypic blocks, and Conjecture 2 predicts the symmetry class of each: $\mathbb{Z}_3$ covers should follow $\widetilde{TW}_2$, quaternion covers $\widetilde{TW}_4$, the regular graph families of section 3 $\widetilde{TW}_1$, and the mixed families $\mathbb{Z}_4$ and $\mathbb{Z}_5$ no single Tracy-Widom law at all.

4.1. Ensemble Selection in the Pure Families. Figure 4.2 is the structural result of this paper: for each pure family, the Kolmogorov-Smirnov distance to the law of its own symmetry class falls to the sampling floor while the distance to $\widetilde{TW}_1$ (dotted) stalls. At the largest sizes, the $\mathbb{Z}_3$ family reaches distance 1.0×10^{-3} from $\widetilde{TW}_2$ (floor at 10^6 samples: roughly 9×10^{-4}) while its distance to $\widetilde{TW}_1$ is stuck at 5.4×10^{-3} , and the quaternion family reaches 7.5×10^{-4} from $\widetilde{TW}_4$ with distance 8.8×10^{-3} to $\widetilde{TW}_1$. The mass to the left of the mean (Figure 4.3) separates the classes at higher resolution: the $\mathbb{Z}_3$ series converges to 0.5145 against the TW_2 value 0.515016, the quaternion series to 0.5109 against the TW_4 value 0.511069, and the regular series to 0.5201 against the TW_1 value 0.519652, each matching its own class, with every competing class excluded by six or more binomial standard errors for the cover families and by three or more for the legacy regular sample. The affine invariant moments agree: Figure 4.4 shows the sample skewness and excess kurtosis of each family converging to the values of its own class, (0.213, 0.093) at the largest $\mathbb{Z}_3$ size against the TW_2 values (0.224, 0.093), and (0.156, 0.042) at the largest quaternion size against the TW_4 values (0.166, 0.049). Figure 4.1 gives the quantile level view of the same selection, plotting the $\mathbb{Z}_3$ sample against the $\beta = 2$ quantiles and the quaternion sample against the $\beta = 4$ quantiles.

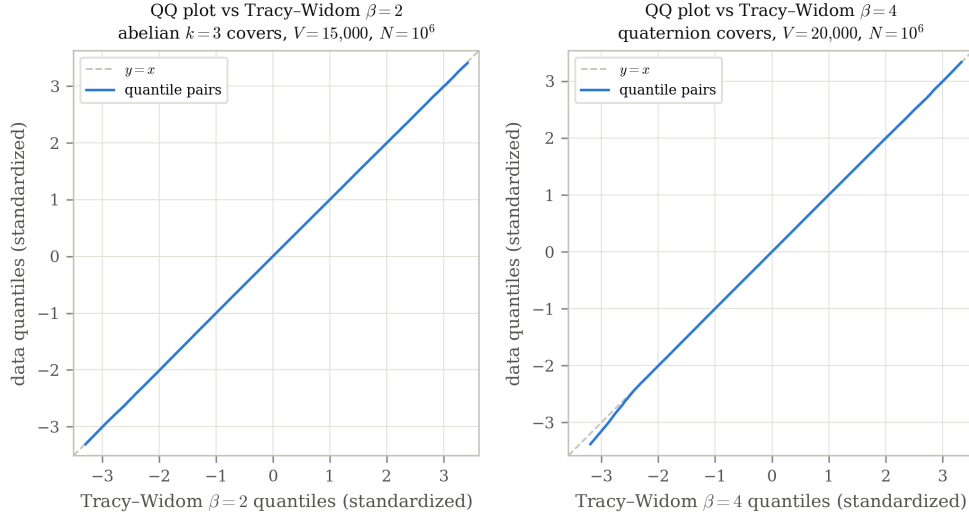


FIGURE 4.1. Quantile-quantile plots of the largest $\mathbb{Z}_3$ cover dataset against Tracy-Widom $\beta = 2$ (left) and of the largest quaternion cover dataset against $\beta = 4$ (right), both standardized, probability range 10^{-4} to 0.999.

Figure 4.5 overlays the standardized densities of the largest datasets from five families on the $\widetilde{TW}_1$ density: at the resolution of the eye all collapse to the Tracy-Widom shape, and the discrimination between the β classes lives in the statistics above.

4.2. The Mixed Families and the Maximum Law. The largest new eigenvalue of a $\mathbb{Z}_4$ cover is the maximum of a GOE class maximum and a GUE class maximum, and that of a $\mathbb{Z}_5$ cover is the maximum of two GUE class maxima, and so under Conjectures 2 and 3 neither should converge to any single Tracy-Widom law. This is visible in the raw statistics: the Kolmogorov-Smirnov distance of the $\mathbb{Z}_4$ family to every candidate law grows with V , reaching 1.9×10^{-2} against $\widetilde{TW}_1$ at the largest size, and the $\mathbb{Z}_5$ family, nearest $\widetilde{TW}_1$ at moderate sizes, drifts away from it as V grows.

The maximum structure can be tested quantitatively. If the two blocks of a family fluctuate independently with the same centering $2\sqrt{d-1}$ and the same scale constant, then the standardized largest new eigenvalue converges to the standardized law of

$$M = \max(X_1, X_2),$$

where X_1, X_2 are independent Tracy-Widom variables of the appropriate classes on a common scale. Computing the distribution function $F_1 F_2$ (for $\mathbb{Z}_4$) and F_2^2 (for $\mathbb{Z}_5$) numerically gives the predicted constants in Table 1: the ratio $|\mu_M|/\sigma_M$, the mass of M left of its mean, and the Ramanujan functional $F_{\beta_1}(0)F_{\beta_2}(0)$. The $\mathbb{Z}_5$ family, whose two blocks have identical marginal distributions and for which the equal scale hypothesis is automatic, matches all three predictions to three decimal places at $n = 5000$. The $\mathbb{Z}_4$ family, whose blocks belong to different classes and converge at visibly different rates, is still drifting at $n = 5000$, in each statistic monotonically toward the predicted value. The maximum structure is visible at the moment level as well: at $n = 5000$ the $\mathbb{Z}_4$ skewness and excess kurtosis are (0.570, 0.658) against the $\max(TW_1, TW_2)$ values (0.584, 0.670), and the $\mathbb{Z}_5$ values are (0.320, 0.212) against (0.329, 0.225), in each case drifting monotonically toward the prediction. Note that the drift away from the pure

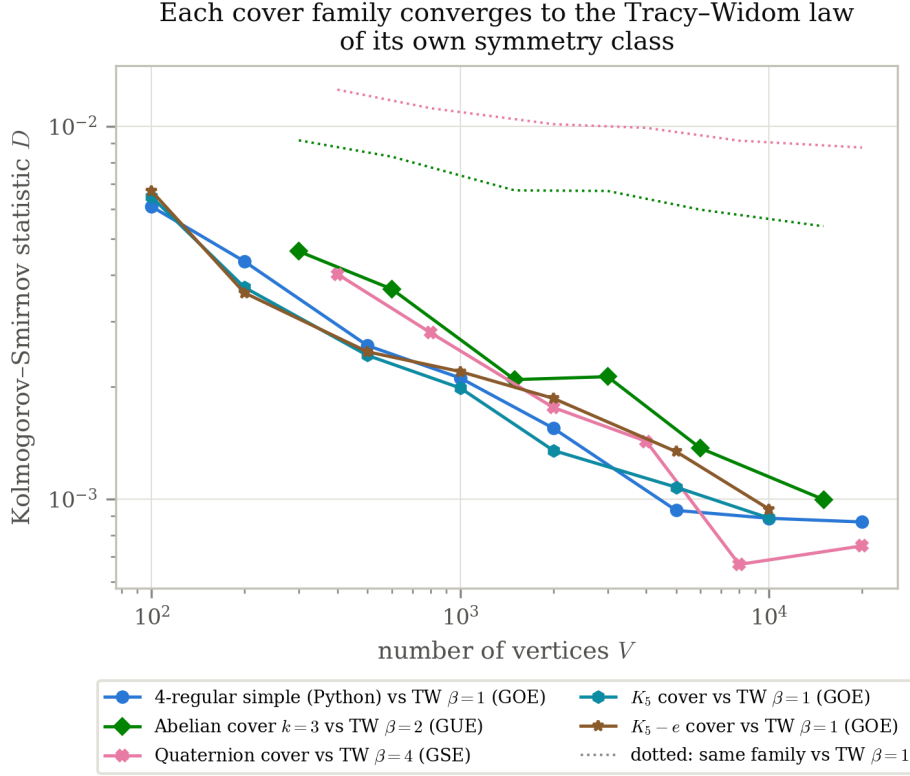


FIGURE 4.2. Kolmogorov-Smirnov distance of each pure family to the Tracy-Widom law of its own symmetry class (solid): regular graphs and the fixed base K_5 and $K_5 - e$ cover families to $\beta = 1$, $\mathbb{Z}_3$ covers to $\beta = 2$, quaternion covers to $\beta = 4$, with the distance to $\beta = 1$ dotted for contrast. The matching class distances fall to the sampling noise floor. The mixed families $\mathbb{Z}_4$ and $\mathbb{Z}_5$ match no single law.

laws is invisible at moderate sizes: with 10^6 samples at $V = 2500$ the Kolmogorov-Smirnov test accepts $\widetilde{TW}_1$ for the $\mathbb{Z}_5$ family ($p = 0.81$), and only the larger sizes reveal that the distance is growing, a caution for experiments at the scale of MNS.

The Ramanujan column of Table 1 is the experimental form of Conjecture 3. Each family approaches the product $\prod F_\beta(0)$ over its blocks: one GUE block gives $F_2(0) = 0.96937\dots$ for $\mathbb{Z}_3$, two give $F_2(0)^2 = 0.93968\dots$ for $\mathbb{Z}_5$, matched by the observed 0.94011 at $n = 5000$, the mixed $\mathbb{Z}_4$ product $F_1(0)F_2(0) = 0.80643\dots$ is approached from above through 0.82293, and the quaternion block gives $F_4(0) = 0.99857\dots$ against the observed 0.99825. Figure 4.6 shows the full curves, and Figure 4.7 the underlying ratios g_V , each family flattening onto the ratio of its own limit law. In particular the limiting probability of being Ramanujan is not one number but a representation theoretic invariant of the family, always strictly between 0 and 1.

4.3. Scaling Amplitudes. All nine series share the $V^{-2/3}$ rate, with family dependent amplitudes (Figures 3.5 and 4.8). The fixed base families are treated in section 5. For the abelian families the amplitudes themselves appear to carry no new constant: each character block A_χ is the adjacency matrix of the random base graph with its entries rotated by the unimodular phases $\chi(\sigma_e)$, a perturbation which does not change the

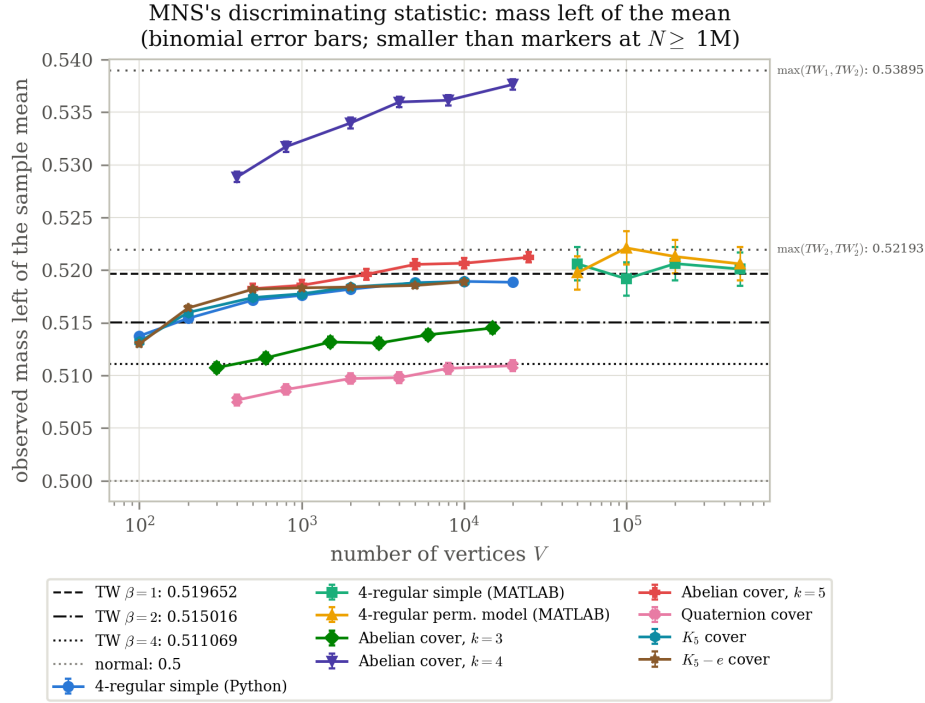


FIGURE 4.3. The fraction of samples below the sample mean against V for all families, with the Tracy-Widom $\beta = 1, 2, 4$ and normal reference values. This replicates MNS's decisive experiment at fifty times their sample size: with 10^6 to 5×10^6 samples the binomial standard error is between 2×10^{-4} and 5×10^{-4} and the four candidate laws are separated by tens of standard errors. Each pure family selects the value of its own symmetry class, while the mixed family $\mathbb{Z}_4$ drifts to ≈ 0.538 , consistent with the maximum law of Table 1 and no Tracy-Widom law.

Third and fourth moments converge to the Tracy-Widom values of each family's symmetry class (permutation model omitted: kurtosis up to $\sim 10^4$)

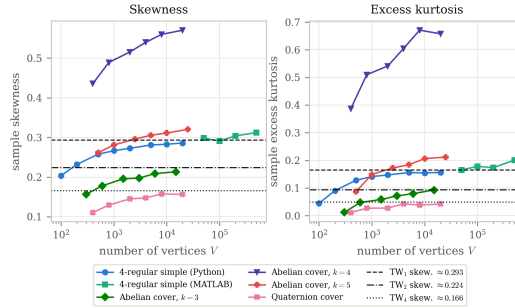


FIGURE 4.4. Sample skewness and excess kurtosis against V with the Tracy-Widom $\beta = 1, 2, 4$ reference values. The regular families converge to the $\beta = 1$ moments, $\mathbb{Z}_3$ to $\beta = 2$, and quaternion to $\beta = 4$. The multigraph model (excess kurtosis up to 10^4) and the fixed base cover families, whose third and fourth moments are dominated by the far outlier tails of section 5, are omitted.

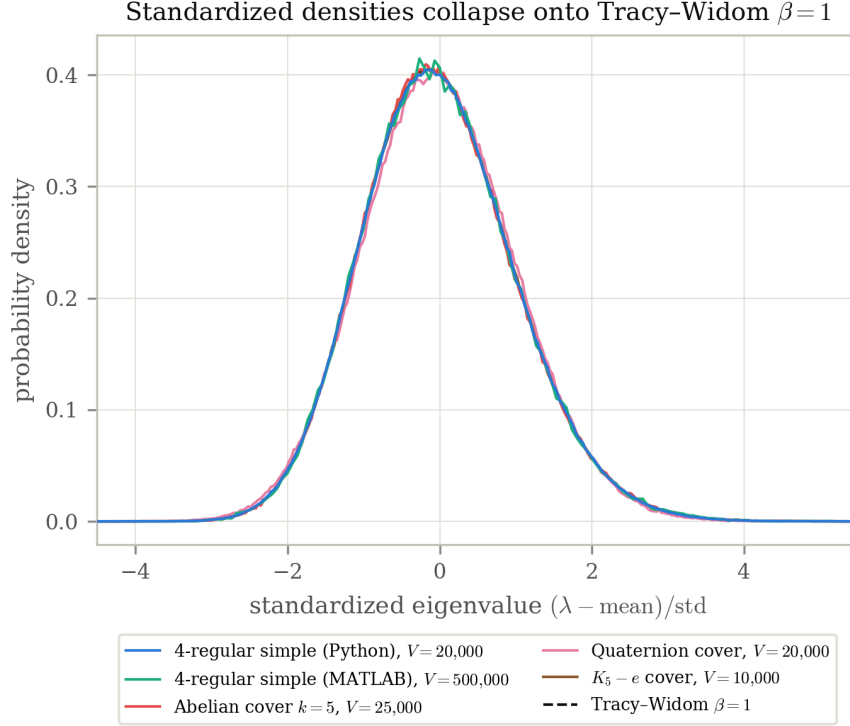


FIGURE 4.5. Standardized empirical densities of the largest datasets from five families overlaid on the standardized $\beta = 1$ Tracy-Widom density. The β classes are indistinguishable at this resolution, and are separated by the statistics of Figures 4.2, 4.3, and 4.4.

family	limit law	g_V	mass left of mean	$P(\text{Ramanujan})$
simple 4-regular	TW_1	0.9993 (0.9515)	0.52011 (0.51965)	0.84352 (0.83191)
$\mathbb{Z}_3$ covers	TW_2	1.9369 (1.9640)	0.51450 (0.51502)	0.96793 (0.96937)
$\mathbb{Z}_4$ covers	$\max(TW_1, TW_2)$	0.8811 (0.8099)	0.53763 (0.53895)	0.82293 (0.80643)
$\mathbb{Z}_5$ covers	$\max(TW_2, TW_2')$	1.6245 (1.6219)	0.52122 (0.52193)	0.94011 (0.93968)
quaternion covers	TW_4	3.1263 (3.2061)	0.51094 (0.51107)	0.99825 (0.99857)
K_5 covers	TW_1	0.9551 (0.9515)	0.51891 (0.51965)	0.83281 (0.83191)
$K_5 - e$ covers	TW_1	0.9387 (0.9515)	0.51890 (0.51965)	0.82916 (0.83191)

TABLE 1. Observed statistics at the largest available size of each family against the values predicted by Conjectures 2, 3, and 4: the ratio $g_V = (\rho - \mu_V)/\sigma_V$ of the mean gap to the standard deviation, the mass left of the sample mean, and the fraction of samples below the Ramanujan threshold ρ ($\rho = 2\sqrt{3}$ except in the last row, where $\rho = \rho(K_5 - e) = 3.2628764\dots$). Predicted values for the maximum laws were computed numerically from the Tracy-Widom distributions assuming independent blocks on a common scale. Each cell shows the observed value with the predicted limit in parentheses. Observed values are at $V = 500,000$ (simple), $n = 5000$ (abelian and quaternion covers), and $V = 10,000$ (the fixed base families).

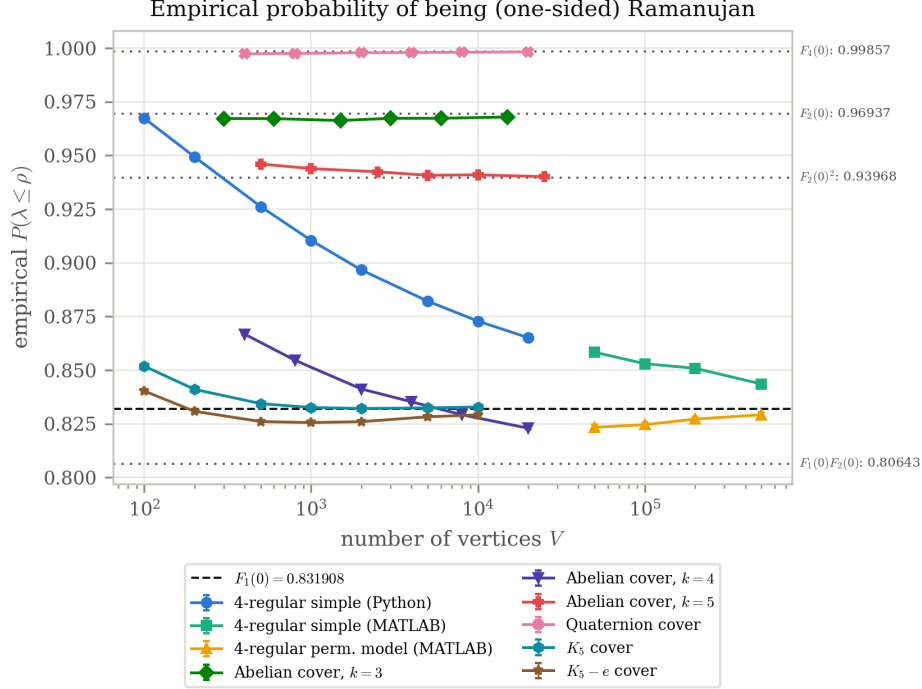


FIGURE 4.6. Empirical $P(\lambda \leq \rho)$ against V for all families, where ρ is the family's Ramanujan threshold, with the limit $F_1(0) = 0.8319081\dots$ for the $\beta = 1$ families. Each cover family approaches the product of Tracy-Widom functionals of Conjecture 3 listed in Table 1.

tree-like local structure, and so we conjecture that every non-trivial character block obeys Conjecture 2 with the same constant $c_\chi = c(d)$ as the unphased regular graph, where $c(d)$ is defined in (1.2) and n is the base size. The data supports this to about one percent at $n = 5000$. For $\mathbb{Z}_3$ the prediction $\sigma \approx \sigma_{TW_2} c(4) n^{-2/3} = 2.568 \times 10^{-3}$ compares to the measured 2.590×10^{-3} , and the predicted mean gap $|\mu_{TW_2}| c(4) n^{-2/3} = 5.044 \times 10^{-3}$ to the measured 5.017×10^{-3} . For $\mathbb{Z}_5$, using the maximum law of Table 1, the predicted standard deviation and gap are 2.218×10^{-3} and 3.598×10^{-3} against the measured 2.230×10^{-3} and 3.623×10^{-3} . The quaternion family plateaus at an amplitude a clean factor below the analogous prediction: the measured compensated standard deviation $\sigma_V n^{2/3}$ drifts toward $0.534 \approx \sigma_{TW_4} c(4) 2^{-1/6}$ rather than $\sigma_{TW_4} c(4) = 0.599$, suggesting that the correct effective size for the quaternionic block differs from n by a fixed power of 2, a bookkeeping familiar from the conventions of the GSE. We do not regard the exponent $-1/6$ as established.

5. COVERS OF AN IRREGULAR BASE GRAPH

Nothing in Conjecture 4 requires the base graph to be regular, and in the irregular case nothing is proven: the largest new eigenvalue should concentrate at the spectral radius $\rho(B)$ of the universal cover, an algebraic number lying between $2\sqrt{\delta - 1}$ and $2\sqrt{\Delta - 1}$, where δ and Δ are the minimum and maximum degrees of B , with TW_1 fluctuations at scale $k^{-2/3}$ and limiting Ramanujan probability $F_1(0)$, universally in the base graph.

Our base is $K_5 - e$, the complete graph on five vertices minus an edge, one of the smallest irregular base graphs of minimum degree three. Its spectrum is $\{1 \pm \sqrt{7}, -1, -1, 0\}$,

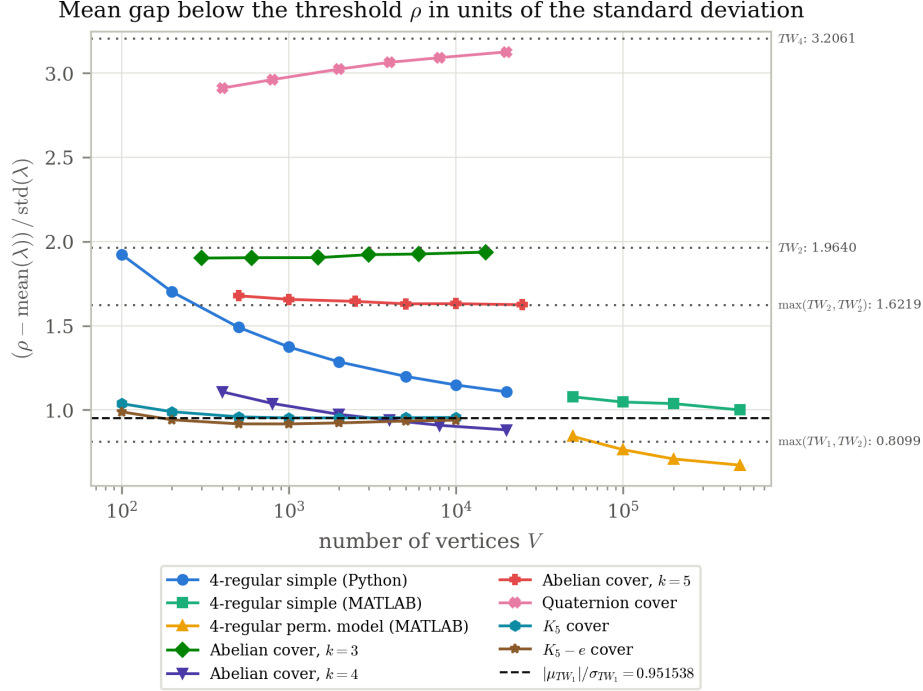


FIGURE 4.7. The ratio $g_V = (\rho - \mu_V) / \sigma_V$ against V for all families, where ρ is the family's Ramanujan threshold, with the $\beta = 1$ reference $|\mu_{TW_1}| / \sigma_{TW_1} = 0.951537\dots$ dashed. MNS conjectured this ratio tends to 0, however each family plateaus at the ratio of its own limit law: 1.96 for $\mathbb{Z}_3$, 1.62 for $\mathbb{Z}_5$, 3.21 for quaternion covers, and the $\beta = 1$ reference for the regular and fixed base families.

with Perron eigenvalue $1 + \sqrt{7} = 3.6457\dots$, and since the largest old eigenvalue exceeds $\rho(K_5 - e)$, the largest new eigenvalue of a cover is not even its second largest eigenvalue, so that the exact removal of the lifted base spectrum matters. By a result of McKay [31], $\rho(K_5 - e)$ is the largest zero of

$$x^{14} - 20x^{12} + 122x^{10} - 152x^8 - 1295x^6 + 4540x^4 - 5948x^2 - 2000,$$

an irreducible polynomial with no expression in radicals, giving

$$\rho(K_5 - e) = 3.26287646593635862827\dots,$$

which we re-derived and verified independently. Note the contrast with the regular case, where $\rho = 2\sqrt{d-1}$: for a general base even the centering of the conjectured law is a non-trivial algebraic quantity (see [5, 31] on computing ρ , and [12] for related spectral experiments on structured random graphs).

Both families were sampled at the scale of the strongest regular graph data: 5×10^6 independent uniform covers per size, at cover degrees $20 \leq k \leq 2000$, giving graphs on $V = 5k$ between 100 and 10,000 vertices. The covers of K_5 are 4-regular graphs with Ramanujan threshold $2\sqrt{3}$, of a different fine structure than the permutation model, since the vertex set carries a fixed partition into five independent fibers with edges only between fibers adjacent in K_5 , and so they test whether the limit law and its constants depend on anything beyond the degree. The covers of $K_5 - e$ test Conjecture 4 where no theorem reaches, around the algebraic centering $\rho(K_5 - e)$.

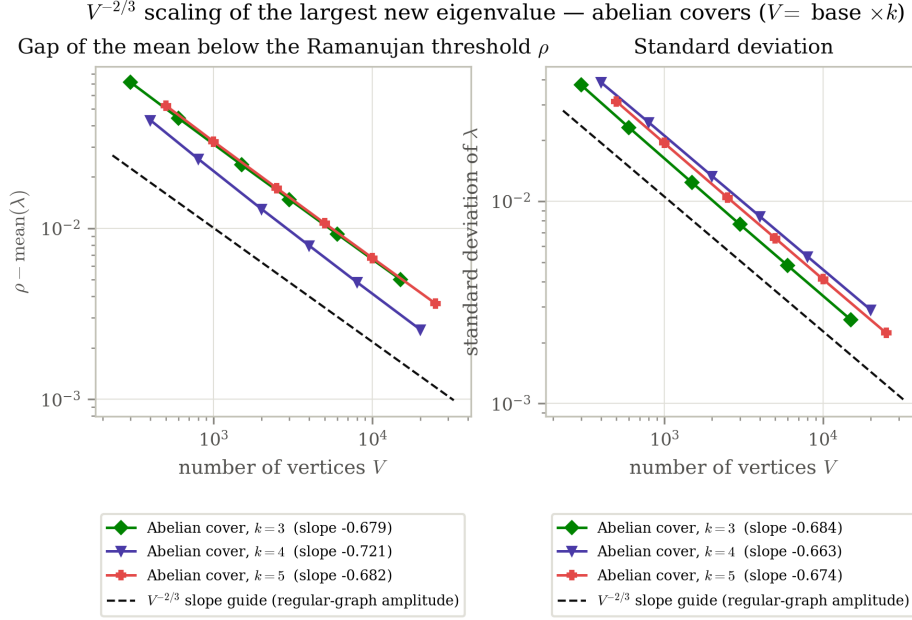


FIGURE 4.8. Mean gap and standard deviation of the largest new eigenvalue against the cover size $V = kn$ for the abelian families, with a $V^{-2/3}$ guide at the regular graph amplitude. The series run parallel to the guide at family dependent offsets.

Figure 5.1 shows the outcome of the distributional test. For both families the Kolmogorov-Smirnov distance to $\widetilde{TW}_1$ falls monotonically with V , reaching 8.9×10^{-4} for K_5 and 9.4×10^{-4} for $K_5 - e$ at $V = 10,000$, within a factor of two of the sampling floor and equal to the simple 4-regular family's distance at the same size, while the distances to the other laws stall or grow, ending at 4.0×10^{-3} ($\widetilde{TW}_2$), 7.9×10^{-3} ($\widetilde{TW}_4$), and 1.9×10^{-2} (normal) for $K_5 - e$. A two sample comparison is sharper still: the Kolmogorov-Smirnov distance between the standardized simple and K_5 samples at $V = 10,000$ is 3.7×10^{-4} ($p = 0.88$), while each lies 8.9×10^{-4} from $\widetilde{TW}_1$ ($p \approx 10^{-3}$), and so at 10^7 combined samples the two models are statistically indistinguishable from one another while both still differ measurably from the limit. At small sizes the families are well separated ($p = 2 \times 10^{-15}$ at $V = 100$), merging from $V \approx 5000$ on: the limit is universal, and the finite size path to it becomes universal as well. Figure 5.2 shows the standardized density of the largest $K_5 - e$ dataset, and Figure 5.3 the corresponding quantile plot. The mass left of the mean reaches 0.51891 (K_5) and 0.51890 ($K_5 - e$) against the TW_1 value 0.519652, on the same convergence path as the simple family at equal sizes, and the ratio g_V reaches 0.9551 and 0.9387 against the TW_1 ratio 0.951537... (Figure 4.7). The local exponents of the mean gap and standard deviation descend to -0.673 and -0.676 for K_5 and -0.670 and -0.678 for $K_5 - e$ over the last pair of sizes, approaching $-2/3$ exactly as for the regular families (Figures 3.7 and 5.4). The empirical Ramanujan probabilities at $V = 10,000$ are

$$P(\lambda_{\text{new}} \leq 2\sqrt{3}) = 0.83281, \quad P(\lambda_{\text{new}} \leq \rho(K_5 - e)) = 0.82916,$$

against the conjectured $F_1(0) = 0.8319081\dots$ (Table 1). Every statistic selects the $\beta = 1$ class, as the reality of the standard representation predicts, and for $K_5 - e$ it does so around a centering which is not $2\sqrt{d-1}$ for any d .

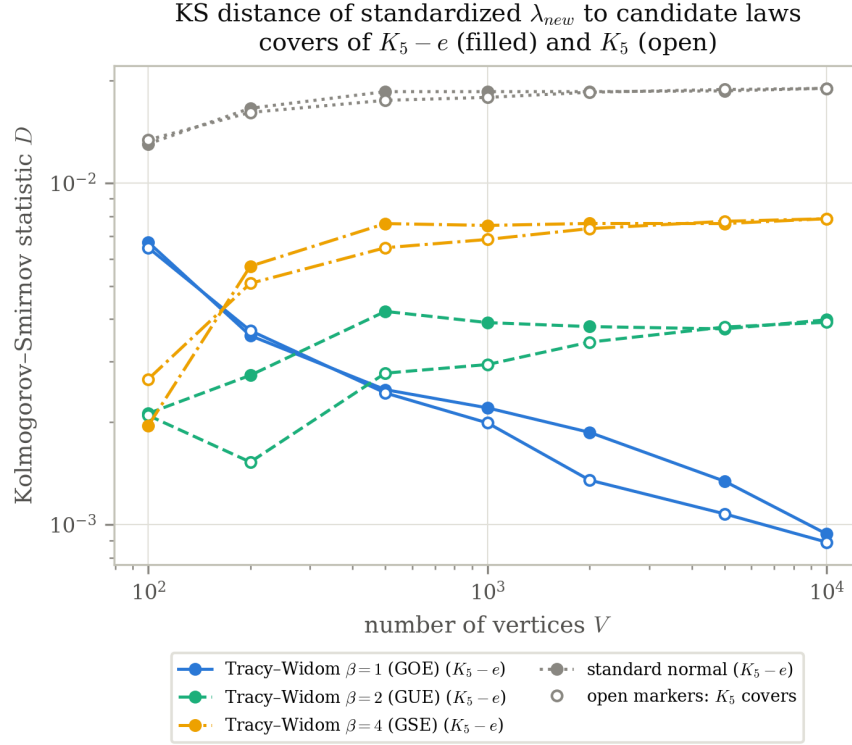


FIGURE 5.1. Kolmogorov-Smirnov distance of the standardized largest new eigenvalue to the four candidate laws for covers of $K_5 - e$ (filled markers, standardized around $\rho(K_5 - e)$) and of K_5 (open markers), 5×10^6 samples per size. For both families only the distance to $\beta = 1$ keeps falling.

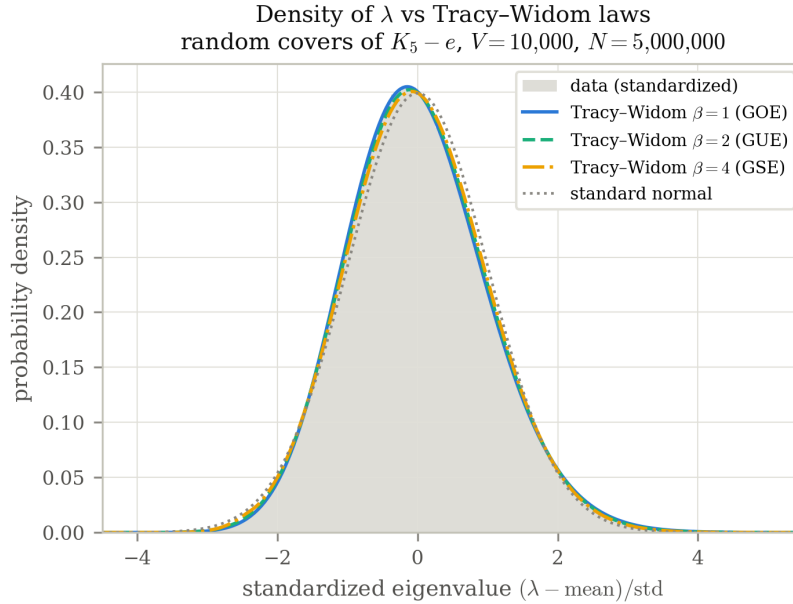


FIGURE 5.2. Standardized empirical density of the largest new eigenvalue over 5×10^6 random covers of $K_5 - e$ at $V = 10,000$, against the standardized Tracy-Widom densities and the standard normal.

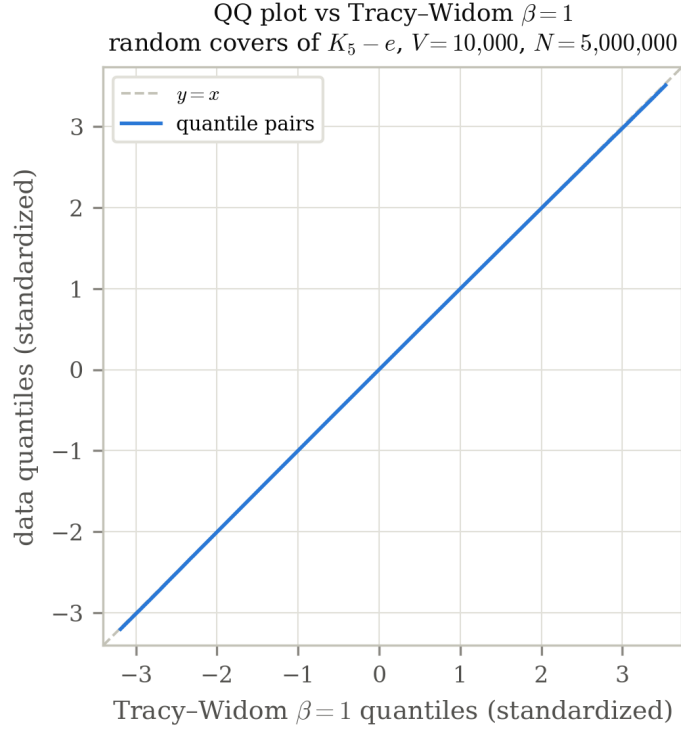


FIGURE 5.3. Quantile-quantile plot of the largest $K_5 - e$ dataset against $\widetilde{TW}_1$, probability range 10^{-4} to 0.999: the bulk of the law around the algebraic centering $\rho(K_5 - e)$ is Tracy-Widom at quantile resolution.

$V^{-2/3}$ scaling of the largest new eigenvalue — covers of K_5 and $K_5 - e$ ($V=5 \times$ cover degree)

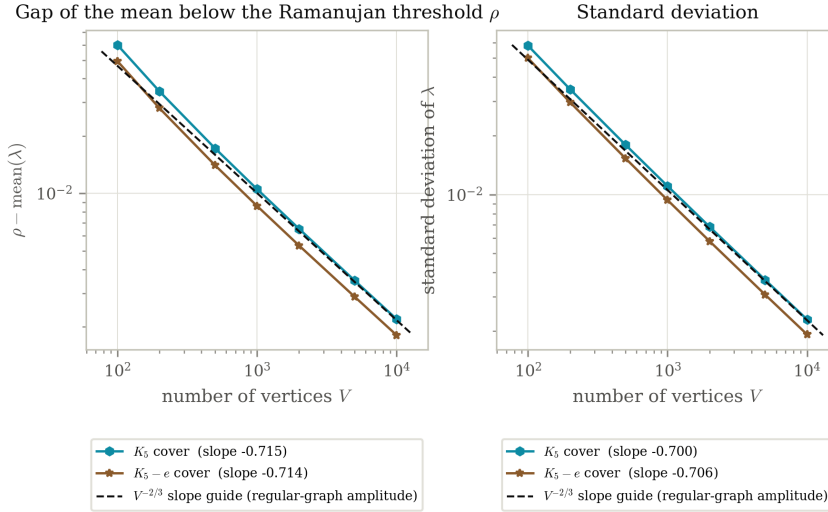


FIGURE 5.4. Mean gap $\rho - \mu_V$ and standard deviation against V for the fixed base families, with a $V^{-2/3}$ guide at the regular graph amplitude. The K_5 series approaches the guide itself, while the $K_5 - e$ series runs parallel to it at a lower amplitude.

Two finer observations. First, the speed of convergence depends strongly on the base graph: the K_5 family sits within 10^{-3} of the limiting Ramanujan probability from $V = 1000$ onward (0.83258 at $V = 1000$), while the permutation model is still at 0.87282 at $V = 10,000$, and the K_5 ratio g_V is already 0.9522 at $V = 1000$ against the permutation model's 1.148 at ten times that size. The limiting constants are universal in the base graph, however the finite size corrections are not, and the fixed base appears to suppress them. In the same direction, the compensated amplitudes of the K_5 family descend to 1.018 and 1.066 at $V = 10,000$, approaching the regular graph constants 1.0047 and 1.0558 of (3.1), and so the amplitude constant of Conjecture 4 for K_5 appears to be $c(4)$ itself, in the convention of (1.2) with V the cover size. The $K_5 - e$ amplitudes plateau lower, with $\sigma_V V^{2/3}$ descending through 0.893 and the compensated gap through 0.839 at $V = 10,000$, and so the $K_5 - e$ analog of $c(d)$ is a new constant, approximately 0.69 against $c(4) = 0.8327$ (the fits $A + BV^{-\beta}$ to the compensated standard deviation and gap give 0.686 and 0.691), the first measurement of an edge scaling constant for an irregular base graph.

Second, the third and fourth moments tell a different story, and the accounting is instructive. The excess kurtosis of the $K_5 - e$ family grows with V , from 0.26 at $V = 100$ to 2.52 at $V = 10,000$ against the TW_1 value 0.165, and its skewness drifts upward through 0.361 against 0.293, while the same statistics for K_5 are 0.28 and 0.292. Figure 5.6 locates the cause in a thin tail of far outliers, at frequencies near 10^{-5} : at $V = 10,000$ a fraction 9×10^{-6} of the $K_5 - e$ samples lie above $\rho + 10\sigma_V$, while the extreme sample of the simple 4-regular family sits only 9 standard deviations out. These outliers are the tangles of [16] made visible sample by sample, and they have exact structure: they accumulate at computable constants (Figure 5.5).

A subgraph of a cover projects to the base graph by a locally injective map, and a fixed connected subgraph H of cycle rank r occurs somewhere in a random degree k cover with probability of order k^{1-r} . When the largest eigenvalue of H , computed with the appropriate branches of the universal cover attached to its vertices, exceeds $\rho(B)$, every occurrence of H seeds a localized eigenvalue at that value. Computing these defect eigenvalues by the standard branch recursion on the universal cover, and confirming them by planting the subgraphs in sampled covers, we have that for $K_5 - e$ no subgraph of cycle rank two binds an eigenvalue above ρ , that K_4 , of cycle rank three, binds

$$\lambda_{K_4} = 3.3391340785 \dots,$$

with its subdivisions filling the interval (ρ, λ_{K_4}) , and that the largest cycle rank four value is 3.4519402..., while a disjoint copy of $K_5 - e$ itself would give $1 + \sqrt{7}$. The data matches this picture in detail. The outlier values at the largest sizes fill an interval above ρ , leave a gap, and cluster on the K_4 atom, where the extreme samples at $V = 5000$ and 10,000 sit, and the extremes at $V \leq 2000$ sit instead on the rank four atom, which decays faster and is gone by $V = 5000$. The frequencies obey the predicted powers: the count in a window around the K_4 atom decays with fitted exponent 2.02 ± 0.13 against the predicted k^{-2} , and at thresholds beyond the reach of cycle rank three the fitted exponent is 2.90 ± 0.14 against k^{-3} . The apparent lengthening of the tail, from 27 standard deviations at $V = 5000$ to 42 at $V = 10,000$, is then exact: a fixed atom measured in units of $\sigma_V \propto V^{-2/3}$ lengthens by $2^{2/3} = 1.587$, and the observed ratio is 1.584. For the K_5 family the same computation gives the atom

$$\lambda_{K_4} = \frac{7}{2}$$

Far outliers lock onto the localized defect eigenvalues (dotted); frequencies decay like k^{1-r}

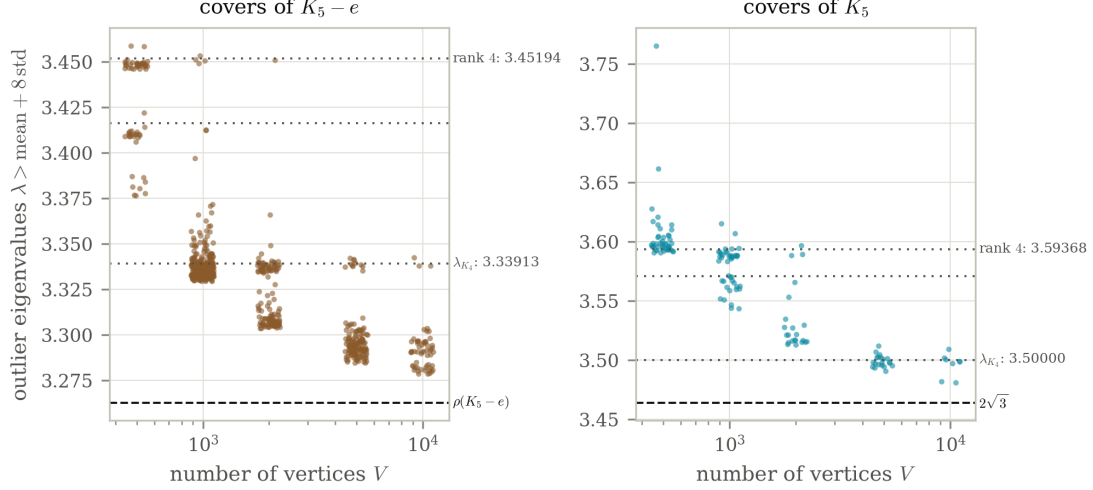


FIGURE 5.5. Every sample more than eight standard deviations above the mean, against V , for the $K_5 - e$ (left) and K_5 (right) cover families, with the computed defect eigenvalues overlaid: the outliers lock onto the atoms $\lambda_{K_4} = 3.33913\dots$ (for K_5 : exactly $7/2$) and the cycle rank four values (dotted), the continuum of K_4 subdivisions drains toward ρ , and the frequencies decay like k^{1-r} .

exactly, and the six samples more than nine standard deviations above the mean at $V = 10,000$ all lie in $[3.497, 3.510]$. For regular Ramanujan bases such as K_5 the polynomial decay of the outlier probability is a theorem of Friedman and Kohler [17], and the atoms with their k^{1-r} frequencies are its microscopic content, and the coexistence of a Tracy-Widom bulk with a thin family of localized outliers is the picture proven at the sparse Erdős-Rényi edge [2, 3, 21]. These outliers are the multigraph phenomenon of section 3.4 in miniature. They dominate the third and fourth moments, which is why the fixed base families are omitted from Figure 4.4, however at frequency 10^{-5} they are invisible to every quantile level statistic, moving the Kolmogorov-Smirnov distance by at most their total mass and the Ramanujan probability by even less. The bulk law is TW_1 , and the atoms ride on top of it.

6. OPEN PROBLEMS

The Block Laws. Conjecture 2 is open in every case, including $\beta = 1$: for covers of a d -regular base of size $n \geq 2$ no analog of Theorem 1 is known, and for irregular bases even the correct formulation of edge rigidity around $\rho(B)$ is open. The abelian case seems the natural first target, since the block A_χ is an explicit random Hermitian matrix built from a random regular graph and iid phases.

Problem 5. Prove that the largest eigenvalue of the $\mathbb{Z}_3$ -character block A_χ of a random $\mathbb{Z}_3$ -cover of a random simple d -regular graph has TW_2 fluctuations, in the sense of Conjecture 2.

The Maximum Laws and the Amplitudes. The mixed families raise a cleaner question than convergence alone: the maximum law depends on the relative scale of the blocks, which we conjectured equal for character blocks in section 4.3.

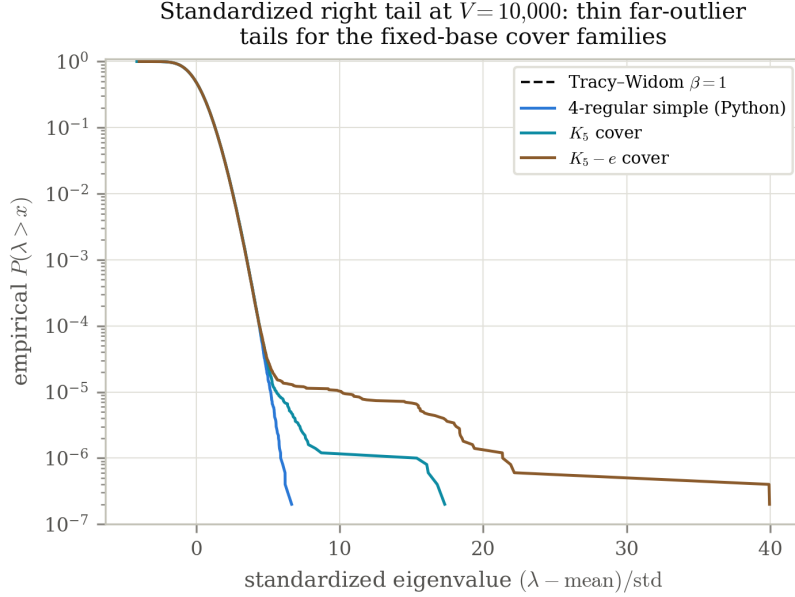


FIGURE 5.6. Standardized right tail $P(\lambda > x)$ at $V = 10,000$ for the simple 4-regular, K_5 cover, and $K_5 - e$ cover families. The bulk of all three collapses onto the Tracy-Widom $\beta = 1$ tail, while the $K_5 - e$ family carries a thin far outlier tail at frequency near 10^{-5} , reaching 42 standard deviations, and the K_5 family a much thinner one.

Problem 6. Determine the scale constants c_ρ of Conjecture 2 and c_B of Conjecture 4. In particular, prove that $c_\chi = c(d)$ for every non-trivial character χ of an abelian cover, determine the quaternionic constant, for which the data suggests $c_\rho = c(d)2^{-1/6}$, and determine c_{K_5-e} , for which the data gives approximately 0.69.

I believe that the equal amplitude prediction $c_\chi = c(d)$ is exactly right, as the one percent discrepancies at $n = 5000$ shrink monotonically with n in our data.

Far Outliers. The permutation multigraph model of section 3.4 converges to the same threshold with the same $V^{-2/3}$ mean gap, however its fluctuations are dominated by localized outliers, and the fixed base families of section 5 carry a much thinner tail of the same character.

Problem 7. Determine the limiting distribution of the largest non-trivial eigenvalue of the permutation multigraph model. Is it a Tracy-Widom law conditioned on the absence of localized eigenvalues, plus an independent point process of outliers?

Problem 8. For a fixed base graph B and $\epsilon > 0$, determine the decay rate of

$$P(\lambda_{\text{new}}(H_k) > \rho(B) + \epsilon)$$

for a uniformly random degree k cover H_k of B . We conjecture that the rate is $k^{1-r(\epsilon)}$ up to constants, where $r(\epsilon)$ is the least cycle rank of a graph mapping locally injectively into B whose defect eigenvalue exceeds $\rho(B) + \epsilon$: the $K_5 - e$ data gives fitted exponents 2.02 ± 0.13 below λ_{K_4} and 2.90 ± 0.14 above it. For regular Ramanujan bases this is essentially the theorem of Friedman and Kohler [17], and the irregular case is open.

Growing Covers. Huang and Yau [23] proved Tracy-Widom fluctuations for regular graphs of growing degree. The analogous regime for covers, base size fixed and cover group growing, is precisely the setting of Conjecture 4, however one may also grow the base: for $\mathbb{Z}_k$ covers with k growing with n , the number of GUE blocks grows, and the largest new eigenvalue becomes the maximum of a growing number of Tracy-Widom variables, suggesting a transition from Tracy-Widom to Gumbel fluctuations. For iterated 2-covers of dense random regular graphs, Liu and Zou [26] prove exactly this maximum of independent Tracy-Widom structure.

Problem 9. For random $\mathbb{Z}_k$ covers of a random d -regular graph on n vertices, determine the joint regimes of (k, n) in which the largest new eigenvalue has Tracy-Widom, and Gumbel, fluctuations.

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